

Decision Tree

Part-1:

⇒

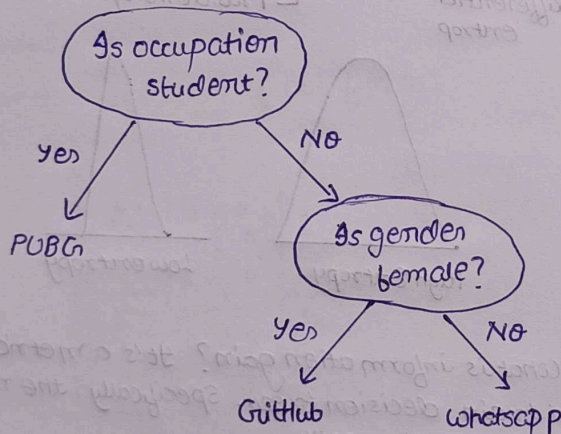
Gender	Occupation	Suggestion
F	Student	PUBG
F	Progra.	GitHub
M	Progra.	Whatsapp
F	Progra.	GitHub
M	Student	PUBG
M	Student	PUBG

if occupation == student
print (PUBG)

else
if gender == female
print (GitHub)
else
print (Whatsapp)

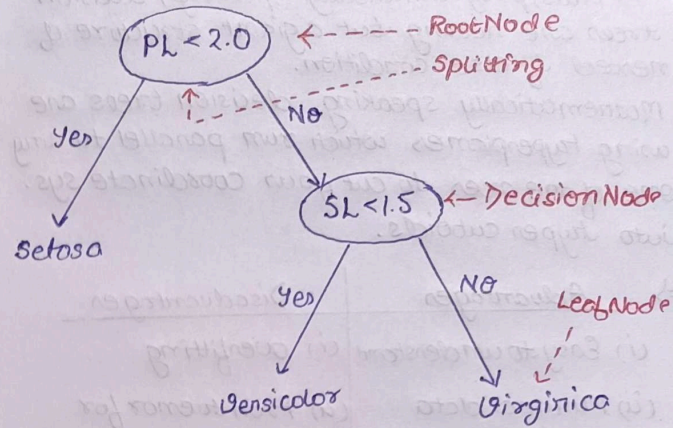
code for such
data shown
above

This can be rep. as tree as:

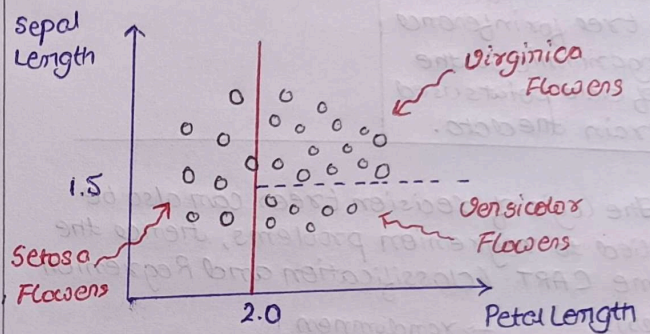


⇒

Petal Length	Sepal Length	Type
1.34	0.34	Setosa
3.45	1.45	Versicolor
1.69	0.98	setosa
2.56	1.79	Virginica
3.00	1.13	Versicolor
1.3	0.88	setosa



This can be shown geometrically as:



⇒ Pseudo code:

- ① Begin with your training dataset, which should have some feature variables and classification or regression output.
- ② Determine the "best feature" in the dataset to split the data on; more on how we define "best feature" later.
- ③ Split the data into subsets that contain the correct values for this best feature. This splitting basically defines a node on the tree, i.e., each node is a splitting point based on a certain feature from our data.
- ④ Recursively generate new tree nodes by using the subset of data created from step 3.

⇒ Thus, programmatically speaking, decision trees are nothing but a giant structure of nested if-else condition.

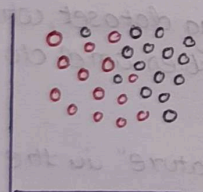
Mathematically speaking, decision trees are using hyperplanes which run parallel to any one of the axes to cut your coordinate sys. into hypercuboids.

Advantages	Disadvantages
(i) Easy to understand	(i) overfitting
(ii) Minimal data preparation is required	(ii) Prone to error for imbalanced datasets.
(iii) The cost of using the tree for inference is logarithmic in the no. of data points used to train the data.	

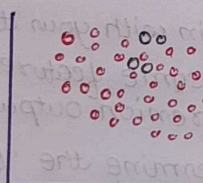
⇒ The logic of decision trees can also be applied to regression problems, hence the name CART (Classification and Regression Trees).

→ randommen

⇒ What is Entropy? More Knowledge I am entropy



(High entropy)



(Low entropy)

⇒ How to calc. entropy?

$$E(S) = \sum_{i=1}^C -P_i \log_2 P_i$$

where, P_i is simply the frequentist prob. of an element/class 'i' in our data.

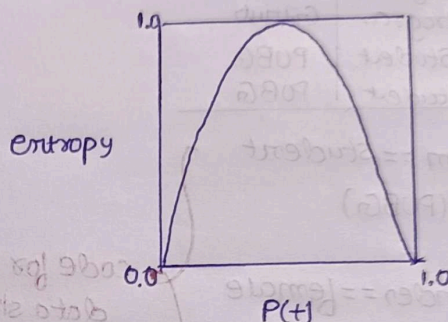
e.g., if our data has only 2 class labels Yes and No,

$$E(Y) = -P_{Yes} \log_2 (P_{Yes}) - P_{No} \log_2 (P_{No})$$

⇒ So we can say,

- More the uncertainty more is entropy.
- For a 2 class problem the min. entropy is 0 and the max. entropy is 1.
- For more than 2 classes the min. entropy is 0 but the max. can be greater than 1.
- Both $\log_2$ or $\log_e$ can be used to calculate entropy.

⇒ Entropy vs Probability:

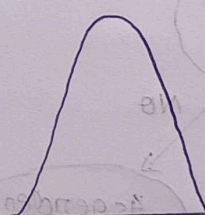


⇒ Entropy for continuous variables:

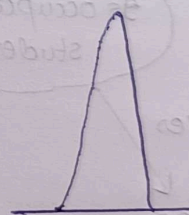
$$h(X) = - \int_{-\infty}^{\infty} f(x) \log f(x) dx$$

↑ differential entropy

↑ Prob. density Function



High Entropy



Low entropy

⇒ What is information gain? It's a metric used to train decision trees. Specifically, the measure the quality of a split.

The information gain is based on the dec. in entropy after a data-set is split on an attribute

Information Gain

$$= E(\text{Parent}) - (\text{Weighted Avg}) * E(\text{Children})$$

⇒ whichever column has the highest info. gain (max decrease in entropy) the algo. will select that column to split the data.

⇒ Decision tree then applies a recursive greedy search algorithm in top-bottom fashion to find info. gain at every level of the tree.

Once a leaf node is reached (Entropy = 0), no more splitting is done.

⇒ Gini Impurity:

$$G_I = 1 - (P_Y^2 + P_N^2)$$

Also, G_I be len for the data where there be len impurity.

e.g., $P_Y = 0.5, P_N = 0.5$

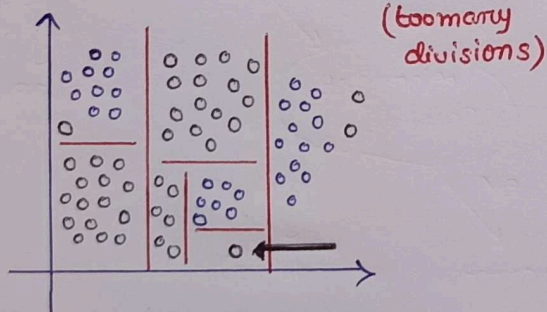
$$\therefore E = 1$$

$$G_I = 1 - (0.5^2 + 0.5^2) = 0.5$$

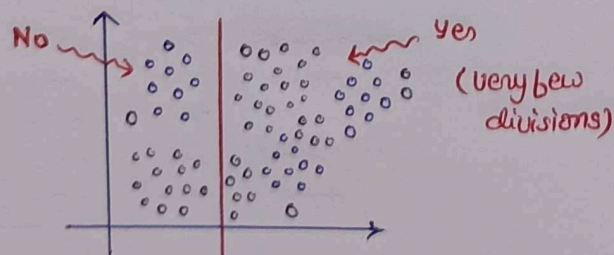
⇒ why Gini exists? It's not so computationally expensive.

Part 2:

⇒ Geometric intuition of Overfitting:

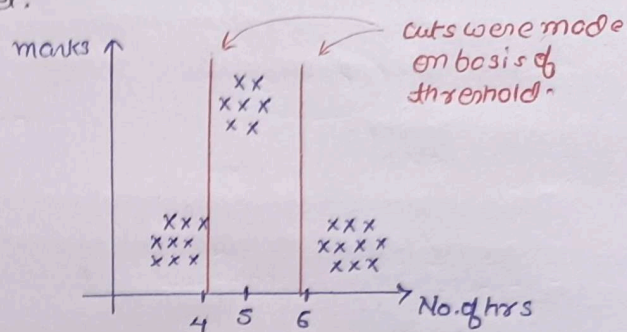


⇒ Geometric intuition of underfitting:

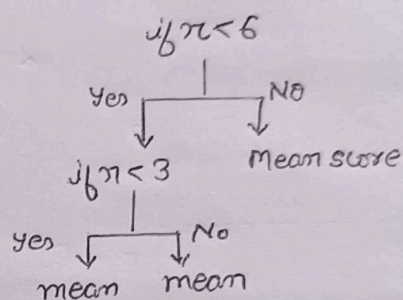


Part - 3:

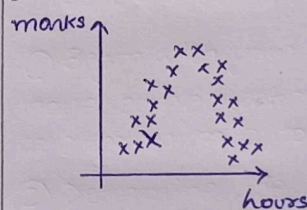
⇒ Regression Tree: It's used for non-linear data set.



So,



⇒ For above example the data was easily separable as thresholds were easy to be predicted. But for any continuous distribution?



No easy predictions of thresholds can be made.